

SAC versus action-conditioned JEPA on the fractal terrain task

RTX 3090 runs, September 20, 2026

Question and result

All runs use the same centered-Gaussian squared-potential terrain, random starts, $P_{\max} = 6$ longitudinal power cap, 16 environments, held-out terrain seeds, and 10,000-transition budget. SAC alone is the strongest observed controller. The normalized, action-conditioned JEPA branch is mechanically valid—it beats shuffled-action and identity diagnostics—but still reduces held-out speed at this budget.

Run	Held-out speed at 10k	Difference from SAC	Status
SAC-only	1.655 ± 0.056	—	baseline
SAC+JEPA, old MSE target	1.278 ± 0.042	−22.8%	superseded cadence/scale safeguards
SAC+JEPA, corrected MSE	1.403 ± 0.020	−15.2%	variance floor and slow EMA
SAC+JEPA, cosine predictive loss	1.188	−28.2%	scale-normalized diagnostic run

Full corrected SAC+JEPA loss

For a replayed transition (o_t, a_t, o_{t+1}) , the actor has online encoder $z_t = f_\theta(o_t)$, EMA target $\bar{z}_{t+1} = \bar{f}(o_{t+1})$, and predictor q_ψ . The optimized actor objective is

$$\begin{aligned} \mathcal{L}_{\text{actor}} = & \mathbb{E}[\alpha \log \pi(a|o) - \min(Q_1(o, a), Q_2(o, a))] \\ & + \lambda (1 - \cos(q_\psi([z_t, a_t]), \text{sg}[\bar{z}_{t+1}])) \\ & + c_{\text{var}} \frac{1}{d} \sum_{j=1}^d \max(0, s_{\min} - \text{std}_B(z_{:,j})). \end{aligned} \tag{1}$$

The cosine loss is scale-free; the variance floor prevents a zero-variance representation. The final run uses $\lambda = 0.025$, $c_{\text{var}} = 0.01$, $s_{\min} = 0.1$, and updates the EMA target by 0.01 every 64 collected transitions. Logged diagnostics include the weighted JEPA and variance terms, action-shuffled loss, identity baseline, and embedding spread.

Interpretation

The cosine replacement fixes the earlier loss-scale failure: the predictive term is no longer numerically negligible relative to the variance term, and the shuffled-action loss exceeds the real-action loss (0.0099 versus 0.0038 at 10k). Thus the predictor uses actions. Nonetheless, performance is lower than SAC alone. At this horizon and budget, the evidence favors SAC-only; multi-step targets and matched multi-seed sweeps remain the next JEPA test.

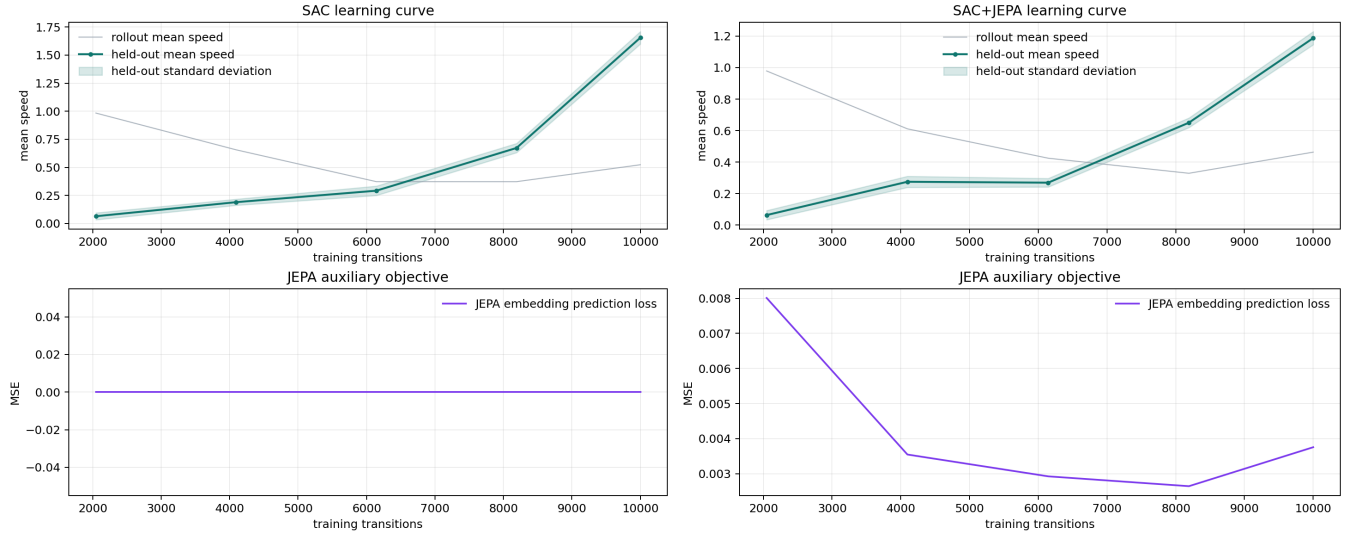


Figure 1: Performance curves. Left: SAC-only. Right: normalized cosine SAC+JEPA. The right lower panel is the scale-free predictive loss.

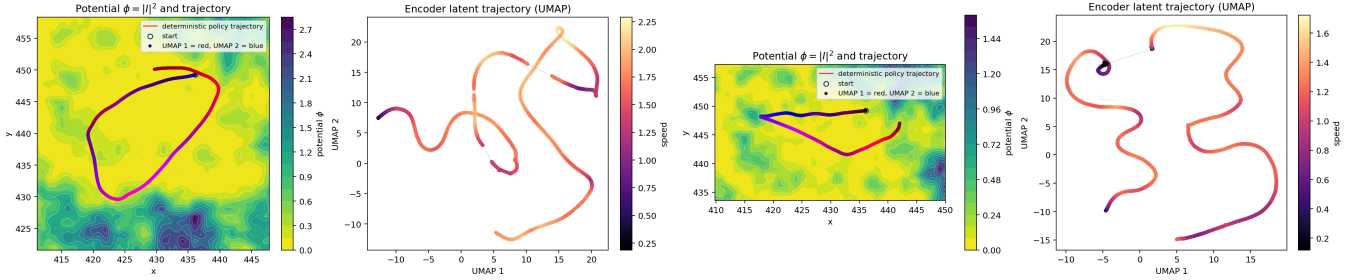


Figure 2: Endpoint terrain-overlay trajectories and UMAPs on the same held-out seed. Left: SAC-only. Right: normalized cosine SAC+JEPA.

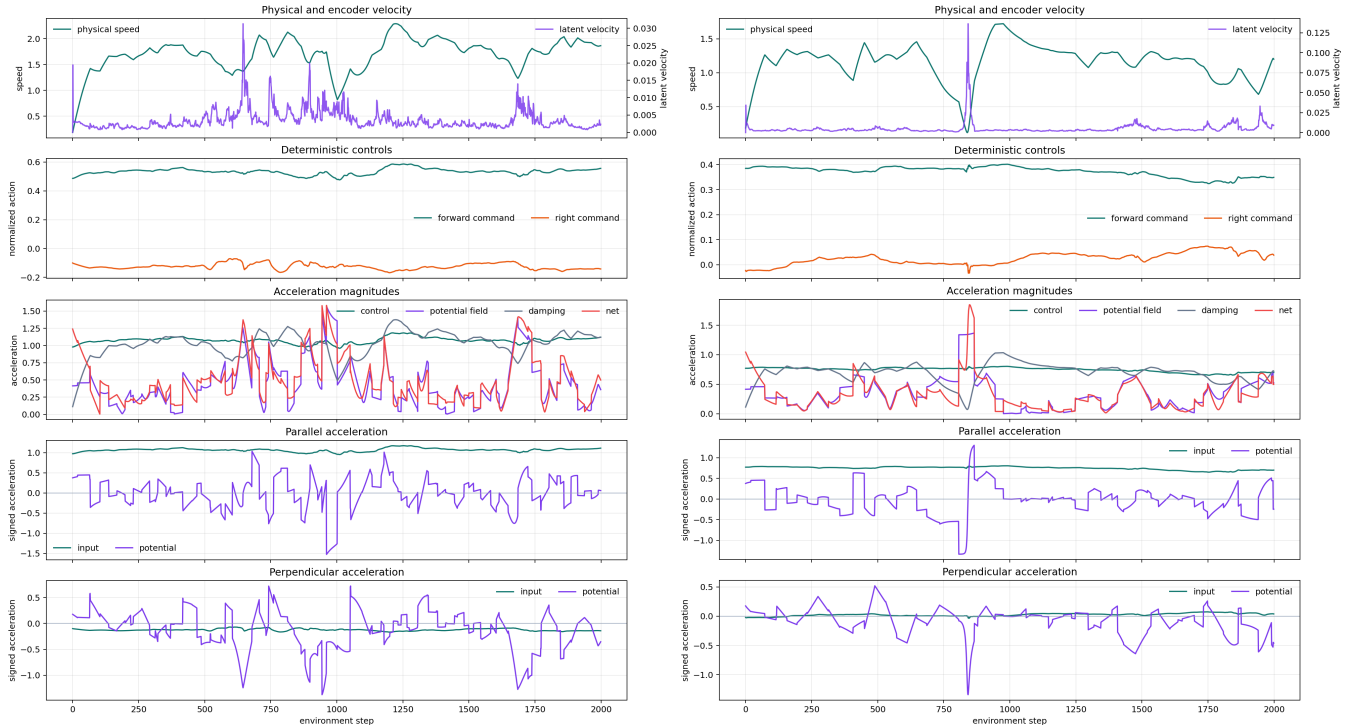


Figure 3: Full velocity, control, and acceleration diagnostics. Left: SAC-only. Right: normalized cosine SAC+JEPA.

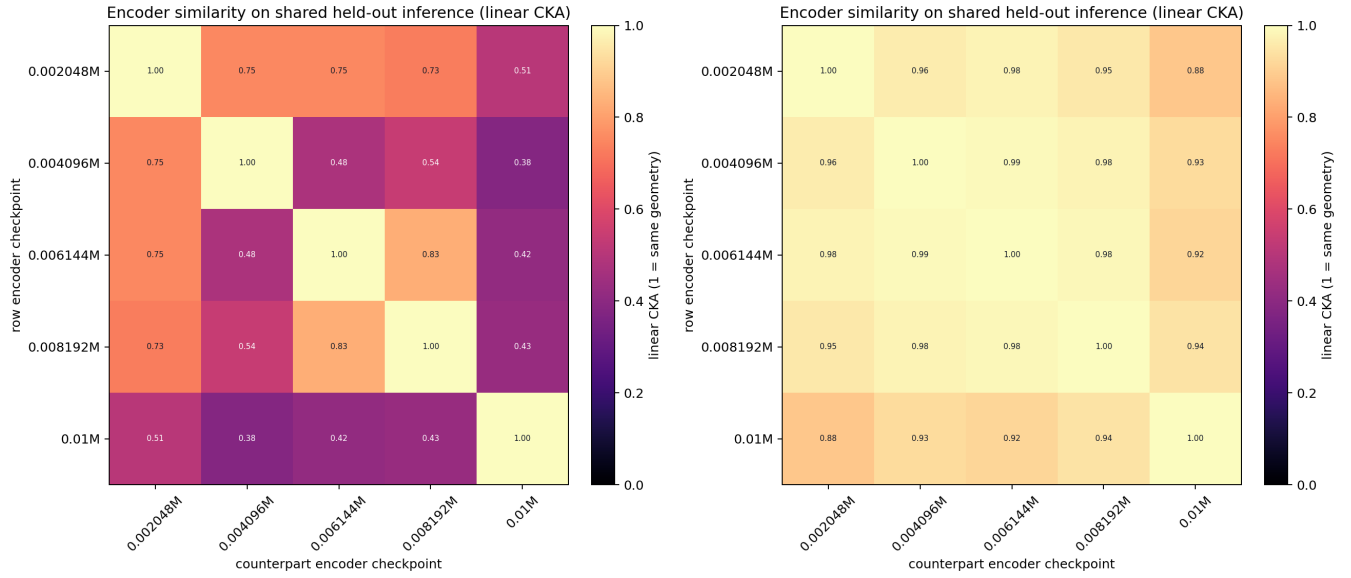


Figure 4: Encoder evolution across the five saved checkpoints, measured by linear CKA on a common held-out inference trajectory.



Figure 5: Checkpoint-pair latent matrix on the shared deterministic held-out trajectory. Each cell jointly embeds the row and counterpart encoder states (blue/orange); diagonal cells are self-comparisons. Left: SAC-only. Right: normalized cosine SAC+JEPA.